Independent FRAMES tuning

Corrected Quiroga-style physical design • noiseless two-qubit rank-one recovery

Frozen setting: 12/12 confirmation trajectories pass the joint gate

Independent tuning changes FRAMES controls only. Quiroga’s published adaptive update has not been reproduced successfully; the exact-line-search FGD result remains a modified diagnostic, not the published competitor.

Declared stage	Audited / planned	Completed	Gate pass	Failures	Unaudited
Screen: 9 configurations × 2 channels	18/18	18	0	0	0
Long stage: frozen promotions	4/4	4	3	0	0
Confirmation: 6 channels × 2 streams	12/12	12	12	0	0

Audited work: 8.2 million updates; 262.4 million charged rows. Fresh-start prefixes included; the 2,000-step timing pilot is excluded.

Shared physical problem and unchanged gate

n=2, d=4; Haar-unitary truth; rank-one factor. All 128 rows: 16 global probes and an 8-outcome POVM with a=b=0.1. Paired normalized Gell-Mann initialization; batch32 and operator-norm radius tau=2.

Joint gate: fidelity >=0.99; full-design RMSE <=1e-6; normalized TP residual <=1e-3. Every component must pass. Numerical failure is never a success.

Method-specific penalty: independently tuned

$$Q_{F,t}(U) = \frac{1}{2m}\|A(UU^H) - y\|_2^2 + \alpha_F(t)H(U)$$

$$\alpha_F(t) = \alpha_0(t+1)^p, \quad \beta_t = [2\alpha_F(t)]^{-1}, \quad p > 0$$

H(U) is the squared, unnormalized TP residual. The initial alpha0 and complete schedule are fixed before a run; the penalty then increases. No shared TP coefficient is imposed on Quiroga/FGD.

Selected configuration: rho-standard. alpha0=5e-05; p=0.25; gamma=min(1,4/(t+4)^1); rho=min(1,4/(t+8)^0.666667).

What was tuned, and what advanced

Predeclared nine-arm screen • channels 0 and 1 • 100,000 updates per trajectory

Rank	Configuration ID	alpha0	p	gamma(t)	rho(t)	Pass	Worst D	Advance
1	rho-standard	5e-05	0.25	4/(t+4)	4/(t+8)^2/3	0/2	2.307	yes
2	penalty-a1-p1	5e-05	0.25	4/(t+4)	2/(t+4)^0.6	0/2	2.428	yes
3	penalty-a2-p0	5e-04	0.125	4/(t+4)	2/(t+4)^0.6	0/2	2.567	—
4	penalty-a1-p0	5e-05	0.125	4/(t+4)	2/(t+4)^0.6	0/2	2.772	—
5	step8	5e-05	0.25	8/(t+8)	2/(t+4)^0.6	0/2	3.148	—
6	step2	5e-05	0.25	2/(t+2)	2/(t+4)^0.6	0/2	3.848	—
7	penalty-a0-p1	5e-06	0.25	4/(t+4)	2/(t+4)^0.6	0/2	3.914	—
8	penalty-a2-p1	5e-04	0.25	4/(t+4)	2/(t+4)^0.6	0/2	4.541	—
9	penalty-a0-p0	5e-06	0.125	4/(t+4)	2/(t+4)^0.6	0/2	39.882	—

gamma and rho are capped at 1. Six arms vary initial penalty and positive growth exponent; three arms change step size or momentum one at a time around the incumbent. The complete candidate grid was frozen before observing its outcomes.

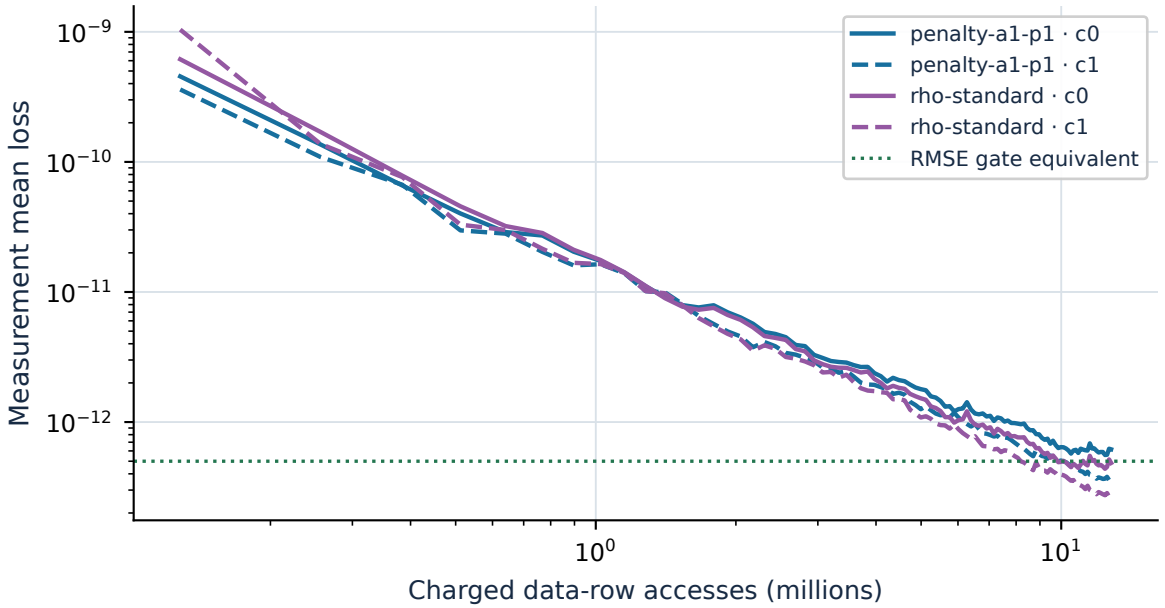
Selection order: fewest numerical failures; most endpoint joint-gate passes; smallest worst-channel deficit D; smallest mean log D; configuration ID. Top two plus the incumbent advance, deduplicated. Every long trajectory starts afresh, including momentum.

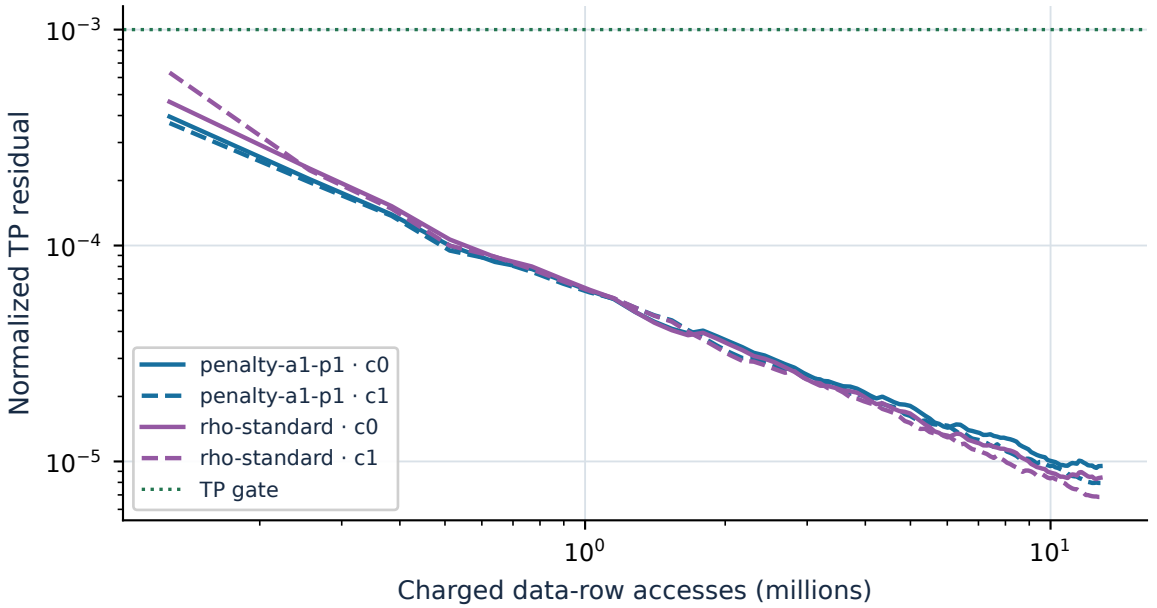
$$D = \max\{\mathrm{RMSE}/10^{-6},\ \mathrm{TP}_{\mathrm{norm}}/10^{-3},\ \max(0, 1-F)/0.01\}$$

100,000 updates × 32 sampled rows = 3.2 million charged rows per trial. Prefix failures are not proofs of nonconvergence.

Full-horizon measurement loss and TP error

Channels 0 and 1 • 400,000 updates per trajectory • identical physical data and initial factors





Configuration	Channel	RMSE	TP normalized	Fidelity	Joint gate	Persistent gate step
rho-standard	0	9.85e-07	8.41e-06	1.0000000000	PASS	400,000
rho-standard	1	7.29e-07	6.85e-06	1.0000000000	PASS	256,000
penalty-a1-p1	0	1.11e-06	9.52e-06	1.0000000000	FAIL	—
penalty-a1-p1	1	8.43e-07	7.95e-06	1.0000000000	PASS	316,000

Mean loss excludes TP penalties; it equals RMSE²/2. Initial factors (zero charged rows) are omitted on logarithmic x axes. Charged rows exclude TP, LMO, metrics and compilation costs, so this is not a wall-time comparison. Persistent means all later recorded checkpoints pass; checkpoints are 4,000 updates apart. A persistent step of 400,000 means only the final recorded checkpoint passed, not demonstrated stable convergence.

Frozen-setting confirmation and limits

Channels 2–7 were exposed in earlier QPT work: confirmation, not an unseen final test

Channel	Stream	RMSE	TP norm.	Fidelity	Gate	Persist. step
2	0	7.68e-07	6.78e-06	1.000000000	PASS	284k
2	1	7.56e-07	6.87e-06	1.000000000	PASS	272k
3	0	8.06e-07	7.16e-06	1.000000000	PASS	276k
3	1	7.17e-07	6.55e-06	1.000000000	PASS	260k
4	0	8.19e-07	7.42e-06	1.000000000	PASS	272k
4	1	7.76e-07	6.87e-06	1.000000000	PASS	260k
5	0	6.70e-07	6.16e-06	1.000000000	PASS	224k
5	1	6.44e-07	5.89e-06	1.000000000	PASS	248k
6	0	6.30e-07	5.81e-06	1.000000000	PASS	240k
6	1	6.65e-07	6.03e-06	1.000000000	PASS	228k
7	0	6.97e-07	6.48e-06	1.000000000	PASS	248k
7	1	7.11e-07	6.62e-06	1.000000000	PASS	256k

12/12 audited trajectories; 12 gate passes; 0 numerical failures. Two streams per channel, one initialization. Sampling streams are not independent channels.

What remains to establish

This is noiseless n=2, rank-one recovery. No noisy, higher-rank or higher-qubit performance follows from these runs.

The JAX CPU dense sensing bridge accelerates local calibration; it is not a large-qubit memory/scaling result. Scan timing includes compilation and is not a controlled head-to-head timing benchmark.

Quiroga’s printed adaptive update remains unresolved. Freeze both independently tuned competitors before final paired tests.

Old-incumbent 400k parity: PASS numerically, not byte-identical across NumPy versions. Max factor difference 9.16e-14; max checkpoint RMSE difference 2.07e-15. The original 1/2 gate outcome is reproduced.

Source 4ce1108a0692: 56 Python files pinned. Screen plan 87c41a281cc8; long selection 64e822315a13. Full source, harness, input-archive and report hashes accompany this PDF.

Final held-out channels 1000–1019 are excluded by every loaded plan; no held-out files are read. Earlier repositories and results are preserved.